According to Newton's third law, the force experienced by the charge q_1 due to q_2 is $\vec{F}_{12} = -\vec{F}_{21}$. Therefore,

$$\vec{F}_{12} = -54\times10^{-3}\,\hat{i}\ \text{N}.$$

The directions of $\vec{F}_{21}$ and $\vec{F}_{12}$ are shown in the above figure in case (a)

(b) $q_1 = +2\ \mu\text{C}$, $q_2 = -3\ \mu\text{C}$, and $r = 1\text{m}$. They are unlike charges. So the force will be attractive.

Force experienced by the charge q_2 due to q_1 is given by

$$\vec{F}_{21} = \frac{9\times10^{9}\times\left(2\times10^{-6}\right)\times\left(-3\times10^{-6}\right)}{1^{2}}\hat{r}_{12}$$
$$= -54\times10^{-3}\text{N}\,\hat{i}\quad (\text{Using } \hat{r}_{12} = \hat{i})$$

The charge q_2 will experience an attractive force towards q_1 which is in the negative x direction.

According to Newton's third law, the force experienced by the charge q_1 due to q_2 is $\vec{F}_{12} = -\vec{F}_{21}$. Therefore,

$$\vec{F}_{12} = 54\times10^{-3}\,\hat{i}\ \text{N}$$

The directions of $\vec{F}_{21}$ and $\vec{F}_{12}$ are shown in the figure (case (b)).

(c) If these two charges are kept inside the water, then the force experienced by q_2 due to q_1

$$\vec{F}_{21}^{W} = \frac{1}{4\pi\epsilon}\frac{q_1q_2}{r^2}\hat{r}_{12}$$

since $\epsilon = \epsilon_r\epsilon_o$

we have $\vec{F}_{21}^{W} = \frac{1}{4\pi\epsilon_r\epsilon_o}\frac{q_1q_2}{r^2}\hat{r}_{12} = \frac{\vec{F}_{21}}{\epsilon_r}$

Therefore,

$$\vec{F}_{21}^{W} = -\frac{54\times10^{-3}\text{N}}{80}\hat{i} = -0.675\times10^{-3}\text{N}\hat{i}$$

Note

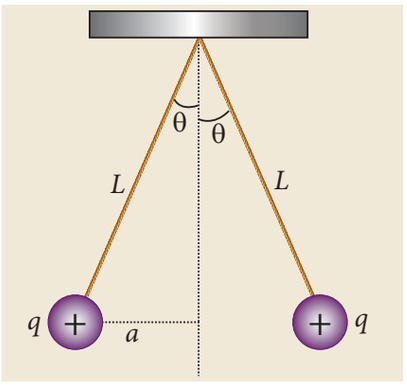
Note that the strength of the force between the two charges in water is reduced by 80 times compared to the force between the same two charges in vacuum.

When common salt (*NaCl*) is taken in water, the electrostatic force between *Na* and *Cl* ions is reduced due to the high relative permittivity of water (ϵ_r = 80). This is the reason water acts as a good solvent.

EXAMPLE 1.3

Two small-sized identical equally charged spheres, each having mass 1 g are hanging in equilibrium as shown in the figure. The length of each string is 10 cm and the angle θ is 30° with the vertical. Calculate the magnitude of the charge in each sphere.

(Take $g = 10\ \text{ms}^{-2}$)



Solution

If the two spheres are neutral, the angle between them will be 0° when

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